CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 12: Reflection

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Find the reflection of $(-2, 5)$ in the x-axis and in the y-axis.

2. Find the reflection of $(4, -7)$ in the origin.

3. Find the reflection of $(3, 6)$ in the line $x = -1$.

4. Find the reflection of $(-2, 1)$ in the line $y = 4$.

5. A point P is reflected in the y-axis to $P'(-3, 2)$. Find P and its reflection in the origin.

6. The image of $B(2, -3)$ in a line parallel to the x-axis is $B'(2, 7)$. Find the equation of the mirror line.

7. Which of the points $(3, 0)$, $(0, -2)$, $(4, 4)$ and $(-5, 0)$ are invariant under reflection in the x-axis?

8. $A(2, 3)$ is reflected in the origin to A', and A' is reflected in the x-axis to A''. Find A' and A'', and a single reflection that maps A to A''.

9. $A(1, 4)$ and $B(5, 4)$ are reflected in the x-axis to A' and B'. Name the figure $ABB'A'$ and find its perimeter.

10. $P(-3, 2)$ is reflected in the line $x = 1$ to Q, and Q is reflected in the line $y = -1$ to R. Find Q and R .

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Answer Key

1. $(-2, -5)$ and $(2, 5)$
2. $(-4, 7)$
3. $(-5, 6)$
4. $(-2, 7)$
5. $P = (3, 2)$; reflection in the origin $(-3, -2)$
6. $y = 2$
7. $(3, 0)$ and $(-5, 0)$
8. $A' = (-2, -3)$, $A'' = (-2, 3)$; reflection in the y-axis
9. $A' = (1, -4)$, $B' = (5, -4)$; a rectangle with perimeter $2(4 + 8) = 24$ units
10. $Q = (5, 2)$, $R = (5, -4)$

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